

Nagizaaz Shaik

GENERATIVE AI ENGINEER · LLM / RAG / AGENTIC SYSTEMS

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PROFESSIONAL SUMMARY

Generative-AI engineer who ships production LLM systems in regulated financial environments. Built an evaluation-gated LLM gateway (Claude / OpenAI / Ollama), RAG systems over regulatory corpora, and async multi-agent research pipelines — with LLMops, cost control, and OWASP LLM Top 10 red-teaming built in. Five years across data and ML engineering, full-time and contract, for financial-services and fintech teams. AWS Machine Learning Specialty certified.

SKILLS

LLM & Agentic AI: RAG (hybrid search, reranking, HyDE, corrective RAG), multi-agent orchestration (LangGraph, CrewAI), Model Context Protocol (MCP), prompt engineering & versioning, semantic caching, LLM gateways, OWASP LLM Top 10, red-teaming

LLMOps & Evaluation: RAGAS, LLM-as-judge, A/B testing, cost analytics, agent observability, MLflow, NVIDIA NIM, vLLM / Ollama

Machine Learning: PyTorch, Transformers (from scratch), XGBoost, BERT, HuggingFace, scikit-learn, SHAP, SageMaker Pipelines & Model Monitor

MLOps & Platform: Apache Airflow, MLflow Registry, SageMaker Pipelines, Docker, Kubernetes, ArgoCD, GitOps, CI/CD, FastAPI

Cloud & Infrastructure: AWS (SageMaker, EC2, S3, RDS, Glue, Redshift, EKS, IAM, CloudWatch, CloudTrail), GCP (BigQuery, Vertex AI), Terraform

Data Governance & Security: NCUA / SOC 2-aware, ECOA / fair-lending testing, PII handling (Presidio), AES-256 (rest / transit), OPA / Conftest, Checkov, Gitleaks

Languages: Python, SQL, PL/SQL, Bash

EXPERIENCE

ML Platform Engineer · Contract

Jul 2025 – Jun 2026

Wright-Patt Credit Union

Fairborn, OH

- Brought in as the credit union's founding ML platform engineer to build its first AWS ML platform end to end — SageMaker inference endpoints, an MLflow model registry with version-gated promotion and automated rollback, and per-environment VPC isolation — cutting deployment failures 85% within the first 60 days.
- Defined the data-security architecture for all ML workloads — AES-256 at rest and in transit, least-privilege IAM, full audit logging — and passed two consecutive NCUA audits with zero findings.
- Built an internal LLM gateway fronting Claude, OpenAI, and Ollama with evaluation guardrails, cost controls, and RAGAS-based grounding checks, letting application teams ship generative-AI features against approved models with measurably better grounded-answer accuracy than the prior single-model setup.
- Codified infrastructure in Terraform and moved deployments to a GitOps workflow on ArgoCD, making environment changes reproducible and peer-reviewed.
- Mentored two engineers into ML and platform ownership through weekly architecture and code-review sessions; both now ship production components independently.

Machine Learning Engineer · Contract

Jul 2024 – Jun 2025

Day Air Credit Union

Kettering, OH

- As Day Air's first dedicated ML engineer, built and deployed an XGBoost credit-risk model on SageMaker scoring 500+ consumer-loan applications per day, automating first-pass underwriting while holding approval-rate stability within risk thresholds.
- Fine-tuned a BERT classifier on member support tickets and shipped it as a FastAPI microservice — 14 intent categories at sub-200ms p99 latency — routing 10,000+ weekly support requests and sharply reducing misrouting.
- Built a retrieval-augmented document-QA system over 500+ internal policy and NCUA regulatory documents with LangChain, vector search, and Claude, cutting compliance research from around two hours to under 20 minutes per query.
- Engineered a weekly retraining pipeline in Apache Airflow with rolling-window retrain, holdout evaluation, and promotion gates on AUC, precision, and calibration, replacing a manual multi-hour process each cycle.
- Deployed drift monitoring via SageMaker Model Monitor with custom statistical baselines, catching two data-drift events before they reached production credit decisions.

Senior Data Engineer

Jul 2020 – Jun 2023

Cognizant (Financial Services client)

Hyderabad, India

- Led an Oracle 11g → Amazon Redshift migration for a financial-services client: redesigned the schema for columnar storage with tuned distribution and sort keys and re-platformed 20+ stored procedures as modular Airflow DAGs, cutting average analytical query runtime 60% across a warehouse serving five business units.
- Designed 15 production ETL/ELT pipelines in Apache Airflow across Oracle, SQL Server, flat files, and REST APIs with a multi-stage validation layer at every hop; surfaced a silent upstream corruption that had been invalidating three months of downstream financial reports.
- Built a real-time Kafka feature pipeline aggregating POS events in five-minute tumbling windows and landing to Redshift at sub-60-second end-to-end latency, supporting a measurable reduction in emergency restocking.
- Promoted from Associate to Senior Data Engineer in two years; led code reviews for three engineers and served as primary client contact across two accounts.

Data / Python Intern (2 rotations)

Jan 2019 – Oct 2019

Uber

Hyderabad, India

- Built SQL and Vertica reporting for driver-utilization, trip-completion, and surge-pricing, and automated a 4-hour weekly ETL process into a scheduled job with Python and pandas — including an anomaly-detection script on rolling 7-day baselines — earning a return-internship offer.

PROJECTS

Finance Agent Crew github.com/shaikn6/finance-agent-crew

Turns a stock ticker into a sourced equity-research brief: an LLM-free gatherer fans out concurrently over SEC EDGAR (XBRL), Alpha Vantage, and news; three analyst agents (fundamental, sentiment, risk) run in parallel; a writer synthesizes a BUY / HOLD / SELL call with bull and bear cases. Non-fatal error handling, CI, Docker.

Python · asyncio · Claude · SEC EDGAR · Alpha Vantage · Pydantic v2 · FastAPI

LLM Safety Auditor github.com/shaikn6/llm-safety-auditor

Reproducible red-teaming harness: 250+ adversarial vectors (50 seed templates × 6 mutation strategies), a 4-layer safety detector, OWASP LLM Top 10 scoring, and compliance-grade PDF reports. Runs API-key-free against a seeded mock LLM or a live provider. 404 tests, 97% coverage; live demo on Hugging Face Spaces.

Python · sentence-transformers · OWASP LLM Top 10 · FastAPI · Streamlit · ReportLab

nano-finbert github.com/shaikn6/nano-finbert

A 1.88M-parameter transformer encoder built entirely from scratch on financial text — custom BPE tokenizer, hand-written multi-head attention, own training loop, no pretrained weights, no HuggingFace dependency (after Karpathy's nanoGPT, every component annotated). Companion fine-tuned model published to Hugging Face at 95.3% held-out accuracy (macro-F1 0.94).

PyTorch · from-scratch Transformer · BPE · NLP · Fintech

EDUCATION

Belhaven University

Jackson, MS

Doctorate of Business Administration (DBA), Applied AI Specialization

Aug 2026 – Present

Focus: applied AI systems, model evaluation, and AI governance for regulated industries.

University of Dayton

Dayton, OH

Master of Science, Computer Science · GPA 3.93 / 4.0

Aug 2023 – May 2025

Coursework: Advanced Machine Learning, NLP, Big Data Systems, Cloud Computing, Distributed Systems, Statistical Inference

CMR Engineering College

Hyderabad, India

Bachelor of Technology, Computer Science Engineering · GPA 8.9 / 10

Aug 2016 – May 2020

CERTIFICATIONS

AWS Certified Machine Learning – Specialty · AWS Certified Cloud Practitioner ·
GCP Professional Data Engineer · Snowflake SnowPro Advanced: Data Engineer ·
dbt Analytics Engineering